

Precision Sensors
for Cleaner Tomorrows



O₂ SENSOR CATALOGUE 2025/2026



OE QUALITY
PRECISION



RELIABLE
PERFORMANCE



LOW EMISSIONS
HIGH EFFICIENCY



iTRA Company (Tehran, Iran)

Basic Specification

Parameter	Specification
Sensor Type	Vehicle-Specific Oxygen Sensor (Siemens Application)
Sensor Technology	4-Wire (Universal Exhaust Gas Oxygen)
Connector	Factory-Fitted Connector
Compatibility	Siemens Applications / ECU Compatible
Primary Function	Air-Fuel Ratio Measurement and Control
Performance Benefits	Optimized Emissions and Improved Fuel Economy
Quality Assurance	100% Factory Tested for Quality and Reliability
Measurement Accuracy	±1%
Recommended torque	42 N.m
Maximum Continuous Operating Temperature	850°C
Maximum Short-Term Temperature	930°C (Maximum 10 Minutes)
Maximum Mounting Torque	45 N.m
Typical Service Life (Unleaded Fuel)	Approximately 500 Hours*
Typical Service Life (Leaded Fuel)	Approximately 50 Hours*

* Sensor lifetime depends on engine operating conditions and application.

Features

- Designed specifically for OEM vehicle applications.
- Supplied with a factory-installed OE connector for plug-and-play installation.
- Optimizes engine emissions and fuel efficiency.
- Extensively tested during manufacturing to ensure maximum quality and reliability.
- Provides accurate wideband lambda measurement for precise engine management.



What Is an Oxygen Sensor?

An oxygen sensor, also known as a lambda sensor, is an electronic device that measures the amount of oxygen remaining in the engine's exhaust gases. Like other types of sensors that detect physical or chemical changes, it converts the measured information into an electrical signal. This signal is transmitted to the vehicle's Electronic Control Unit (ECU), allowing the engine management system to continuously monitor combustion conditions and make precise adjustments to fuel delivery. By maintaining the correct air-fuel mixture, the ECU helps ensure efficient engine operation under varying driving conditions.

Why Are Oxygen Sensors Important?

As environmental regulations become increasingly strict, vehicle manufacturers must minimize harmful exhaust emissions while maintaining fuel efficiency and engine performance. Oxygen sensors play a critical role in achieving these objectives by providing real-time feedback on the combustion process. Using this information, the ECU can optimize the air-fuel ratio to reduce pollutant emissions, improve fuel economy, and enhance overall engine performance. Consequently, oxygen sensors have become a standard component in virtually all modern gasoline-powered vehicles equipped with electronic fuel injection systems.

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How Does an Oxygen Sensor Work?

Modern gasoline engines typically rely on a three-way catalytic converter to reduce harmful exhaust emissions. This catalytic converter transforms hazardous gases—including carbon monoxide (CO), nitrogen oxides (NO_x), and unburned hydrocarbons (HC)—into less harmful substances such as carbon dioxide (CO₂), water vapor (H₂O), and nitrogen (N₂). For the catalytic converter to perform efficiently, the engine must maintain an air-fuel ratio that remains within a very narrow operating range.

The oxygen sensor, positioned in the exhaust system ahead of the catalytic converter, continuously monitors the oxygen concentration in the exhaust stream. Based on its measurements, it sends an electrical signal to the ECU, which instantly adjusts the amount of fuel injected into the engine whenever necessary. This continuous feedback process, known as a **closed-loop control system**, enables the engine to maintain optimal combustion, maximize catalytic converter efficiency, reduce emissions, and improve overall vehicle performance.



When Should an Oxygen Sensor Be Replaced?

Oxygen sensors operate in extremely demanding conditions, where they are continuously exposed to high temperatures, vibration, moisture, and exhaust contaminants. Over time, these factors gradually reduce sensor accuracy and responsiveness. Since driving habits and operating conditions vary significantly from one vehicle to another, the exact service life of an oxygen sensor cannot be universally defined. Vehicles that are frequently driven on short urban trips generally experience faster sensor deterioration than those used primarily for long-distance highway driving.

To ensure reliable engine operation and emission control, it is recommended that the oxygen sensor be inspected every **20,000 miles (approximately 32,000 km)** or at least once a year. A degraded or malfunctioning sensor may cause poor engine performance, increased fuel consumption, higher exhaust emissions, rough idling, or illumination of the **Check Engine** warning light. Modern engine management systems can also record diagnostic trouble codes (DTCs) related to oxygen sensor performance, making fault identification easier. If left unrepaired, a defective sensor may eventually damage the catalytic converter, resulting in significantly higher repair costs.

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Any sensor that has suffered physical damage—including damage to the sensing element, wiring harness, or electrical connector—should be replaced immediately, as its performance and reliability can no longer be guaranteed.

How to Test an Oxygen Sensor

Testing should always begin with a thorough visual inspection of the sensor, its wiring, and the electrical connector. Before performing any inspection, switch off the engine and allow the exhaust system to cool completely to avoid burns or injury. Look for signs of corrosion, broken wires, melted insulation, contamination, or mechanical damage that could affect sensor operation.

For heated oxygen sensors, which typically have **three or more wires**, the condition of the internal heater circuit should also be verified using an ohmmeter. Care must be taken to identify the correct heater terminals before taking measurements. Under normal conditions, the heater element should exhibit a finite resistance within the manufacturer's specified range. A resistance value **greater than 30 Ω** generally indicates a faulty heater, while an **infinite (∞) resistance** reading signifies an open circuit caused by a broken heater element or damaged internal wiring. In either case, the oxygen sensor should be replaced to restore proper engine management and emission control.



Why Choose Vehicle-Specific Oxygen Sensors?

Vehicle-specific oxygen sensors are designed to match the exact requirements of each engine application, ensuring reliable performance and long-term durability. Unlike universal sensors, they are manufactured to meet or exceed original equipment (OE) specifications, providing precise compatibility with the vehicle's engine management system.

Key Benefits

- **OE-Level Compatibility** – Manufactured to meet or exceed original equipment specifications for fit, performance, and reliability.
- **Application-Specific Design** – Sensor dimensions, insertion depth, and protective shield configuration are engineered specifically for each vehicle model.
- **Direct Plug-and-Play Installation** – Equipped with the correct factory-style connector, eliminating the need for wire cutting or splicing.
- **Eliminates Wiring Errors** – Dedicated connectors prevent incorrect electrical connections during installation.
- **Reliable Electrical Performance** – Factory-matched connectors reduce the risk of high-resistance contacts, corrosion, and moisture intrusion commonly associated with universal sensors.
- **Improved Connection Integrity** – Pre-assembled connectors help avoid poor electrical joints that may develop over time with manually spliced wiring.
- **Faster Installation** – Direct-fit construction significantly reduces installation time and simplifies replacement.
- **Proven Quality** – Manufactured under stringent quality standards by one of the world's leading producers of oxygen sensors.
- **Certified Performance** – Independently tested and certified for compatibility with original equipment requirements by recognized technical certification authorities.

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Installation Guidelines

Important: Always consult the vehicle manufacturer's service manual for model-specific installation procedures and torque specifications.

1. Allow the engine and exhaust system to cool completely before beginning work. Disconnect the battery if recommended by the vehicle manufacturer. Remove the old oxygen sensor using the appropriate sensor socket or removal tool.
2. Inspect the vehicle wiring harness and electrical connector for signs of corrosion, wear, or physical damage. Repair or replace damaged components before installing the new sensor.
3. Ensure that the threaded mounting hole in the exhaust system is clean and undamaged to allow proper installation.



4. Verify that the replacement oxygen sensor matches the correct part number for the vehicle application.
5. Handle the new sensor carefully. Avoid dropping or striking it, as impact may damage the internal ceramic sensing element without leaving visible external signs.
6. Most new sensors are supplied with a factory-applied anti-seize coating on the threads. Do **not** apply additional anti-seize compounds unless specifically instructed by the manufacturer, as some products may contaminate or damage the sensing element.
7. Avoid exposing the sensor to cleaning chemicals, lubricants, or anti-rust sprays, since these substances can adversely affect sensor performance.
8. Compare the cable length of the replacement sensor with the original. If adjustment is required, follow the installation instructions supplied with the product.
9. Thread the sensor into the exhaust by hand to prevent cross-threading, then tighten it to the manufacturer's specified torque using the appropriate tool. Ensure the wiring is not twisted during installation.
10. Secure the wiring harness using the original clips or fasteners. Route the cable away from the exhaust system, moving components, and sharp edges while avoiding excessive bending or tension.
11. Reconnect the electrical connector and, if disconnected, reconnect the battery. Clear any stored diagnostic trouble codes (DTCs) according to the vehicle manufacturer's recommended procedure, then verify proper system operation.

Recommended Tightening Torque

To ensure proper sealing and prevent thread damage, install the oxygen sensor using the manufacturer's specified tightening torque.

THREAD SIZE	RECOMMENDED TORQUE
M18 × 1.5 MM	35–45 N·m
M12 × 1.25 MM	18–23 N·m

Note: Always start threading the sensor by hand to avoid cross-threading, and use a calibrated torque wrench to achieve the specified tightening torque.



THE NO.1 CHOICE THE WORLD OVER



iTRA

Precise Details, Idea Realization

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